

JKBOSE CLASS 10TH - MATHEMATICS

Series: 10th ARM(SZ) JKUT 2024 - Series A-3-X | Code: 1003-X | Total Questions: 40

Time: 3 Hours

Max Marks: 80 Marks

GENERAL INSTRUCTIONS: Annual Regular Soft Zone Examination. Section A (1m), B (2m), C (3m), D (4m).**SECTION - A : OBJECTIVE & VERY SHORT QUESTIONS**

(1 Mark Each)

1. Which of the following numbers is an irrational number? [1 Mark]
(A) 0.101
(B) 0.202202220...
(C) 0.1011
(D) None of these
2. The sum of zeroes of the polynomial $2x^2 - 8x + 6$ is: [1 Mark]
(A) -4
(B) 3
(C) 4
(D) None of these
3. The pair of linear equations $2x + 3y - 9 = 0$ and $4x + 6y - 18 = 0$ are: [1 Mark]
(A) Coincident
(B) Parallel
(C) Intersecting
(D) None of these
4. The 35th term of the A.P. : 21, 18, 15, ... is: [1 Mark]
(A) -18
(B) -71
(C) -81
(D) None of these
5. $\cot 30^\circ$ is equal to: [1 Mark]
(A) $\sec 30^\circ$
(B) $\cot 60^\circ$
(C) $\tan 60^\circ$
(D) None of these
6. The distance of a point from the X-axis is called its: [1 Mark]
(A) Abscissa
(B) Ordinate
(C) Coordinate
(D) None of these
7. H.C.F. of 7 and 28 is: [1 Mark]
(A) 4
(B) 28
(C) 7
(D) None of these
8. The probability that it will rain tomorrow is 0.85. What is the probability that it will not rain tomorrow? [1 Mark]

- (A) 0.25
- (B) 0.145
- (C) 0.05
- (D) None of these

9. The volume of a right circular cone is given by: [1 Mark]

- (A) $\pi * r^2 * h$
- (B) $(1/3) \pi * r^2 * h$
- (C) $3 \pi * r^2 * h$
- (D) None of these

10. Discriminant of the quadratic equation $3x^2 - 2\sqrt{6}x + 2 = 0$ is: [1 Mark]

- (A) 24
- (B) 6
- (C) 0
- (D) None of these

11. The prime factorization of 1771 is $7 * 11 * 13$. (State True or False) [1 Mark]

12. The sum of the first 51 positive integers is: [1 Mark]

- (A) 1316
- (B) 1326
- (C) 1362
- (D) None of these

13. -0.5 can be the probability of an event. (State True or False) [1 Mark]

14. All circles are (Congruent / Similar) [1 Mark]

15. The number of tangents that can be drawn from a point outside the circle is [1 Mark]

16. Write the general formula for the sum to n terms of an A.P. [1 Mark]

17. $x = 0, y = 2$ is a solution of equation $y - 2x = -2$. (State True or False) [1 Mark]

18. The value of $\cos A$ exceeds 1. (State True or False) [1 Mark]

--- OR ---

Fill in the blank: $\sin^2 A + \dots = 1$ (for $0 \text{ deg} \leq A \leq 90 \text{ deg}$).

19. Calculate the mean of the first 6 prime numbers. [1 Mark]

20. Write the formula for the class mark of grouped data. [1 Mark]

--- OR ---

If mean = 15.6 and median = 15.73, then mode =

SECTION - B : SHORT ANSWER QUESTIONS

(2 Marks Each)

21. Solve the pair of linear equations $3x - 5y - 4 = 0$ and $9x = 2y + 7$ by substitution method. [2 Marks]

22. Find the roots of the quadratic equation $2x^2 + x - 6 = 0$ by factorisation. [2 Marks]

23. Given $\sec \theta = 13/12$, calculate all other trigonometric ratios. [2 Marks]

24. Find the volume of a sphere of radius 3 cm. [2 Marks]

--- OR ---

Calculate the curved surface area of a cylinder of radius 2 cm and height 7 cm.

25. Find the value of y for which the distance between the points $P(2, -3)$ and $Q(10, y)$ is 10 units. [2 Marks]

--- OR ---

Check whether $(5, -2)$, $(6, 4)$ and $(7, -2)$ are the vertices of an isosceles triangle.

26. Find a quadratic polynomial, the sum and product of whose zeroes are $\sqrt{2}$ and $1/3$, respectively. [2 Marks]

SECTION - C : SHORT ANSWER QUESTIONS

(3 Marks Each)

27. Find the coordinates of the points which divide the line segment joining $A(-2, 2)$ and $B(2, 8)$ into four equal parts. [3 Marks]

28. Find the area of a quadrant of a circle whose circumference is 22 cm. [3 Marks]

29. Prove that opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle. [3 Marks]

--- OR ---

Two tangents TP and TQ are drawn to a circle with centre O from an external point T . Prove that $\angle PTQ = 2 \times \angle OPQ$.

30. E is a point on the side AD produced of a parallelogram $ABCD$ and BE intersects CD at F . Show that $\triangle ABE \sim \triangle CFB$. [3 Marks]

31. The diagonals of a quadrilateral $ABCD$ intersect each other at point O such that $AO / BO = CO / DO$. Show that $ABCD$ is a trapezium. [3 Marks]

32. Prove that $6 + \sqrt{2}$ is an irrational number. [3 Marks]

33. An A.P. consists of 50 terms of which the 3rd term is 12 and the last term is 106. Find the 29th term. [3 Marks]

--- OR ---

Find the sum of the first 15 multiples of 8.

34. One card is drawn from a well-shuffled deck of 52 cards. Find the probability of getting: (i) A face card, (ii) A spade. [3 Marks]

SECTION - D : LONG ANSWER QUESTIONS

(4 Marks Each)

35. A train travels a distance of 480 km at a uniform speed. If the speed had been 8 km/h less, then it would have taken 3 hours more to cover the same distance. Find the speed of the train.

[4 Marks]

--- OR ---

Find the value of K so that the quadratic equation $Kx(x - 2) + 6 = 0$ has two equal real roots.

36. A solid iron pole consists of a cylinder of height 220 cm and base diameter 24 cm, which is surmounted by another cylinder of height 60 cm and radius 8 cm. Find the mass of the pole, given that 1 cm³ of iron has approximately 8 g mass. (Take $\pi = 3.14$).

[4 Marks]

--- OR ---

From a solid cylinder whose height is 2.4 cm and diameter is 1.4 cm, a conical cavity of the same height and diameter is hollowed out. Find the total surface area of the remaining solid to the nearest cm².

37. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60 deg and the angle of depression of its foot is 45 deg. Determine the height of the tower.

[4 Marks]

38. Evaluate the trigonometric value: $(5 \cos^2 60^\circ + 4 \sec^2 30^\circ - \tan^2 45^\circ) / (\sin^2 30^\circ + \cos^2 30^\circ)$.

[4 Marks]

--- OR ---

Prove the trigonometric identity: $(\sin \theta - 2 \sin^3 \theta) / (2 \cos^3 \theta - \cos \theta) = \tan \theta$.

39. If a line intersects sides AB and AC of a triangle ABC at D and E respectively and is parallel to BC, prove that $AD / AB = AE / AC$.

[4 Marks]

--- OR ---

A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.

40. If the median of the distribution given below is 28.5, find the missing frequencies x and y (Total Frequency = 60):

[4 Marks]

Class Interval	Frequency
0 - 10	5
10 - 20	x
20 - 30	20
30 - 40	15
40 - 50	y
50 - 60	5
Total	60